# FPPF HYPERCOVERINGS: A^{-1} ON ETALE SHEAVES IS FULLY FAITHFUL WITH ESSENTIAL IMAGE THE CARTESIAN SHEAVES

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the fully-faithful half is obtained by what the author calls 'a formal argument (chasing around the diagram)' (source line 7331) through the comparison diagram of lemma-compare-simplicial-objects-fppf-etale - all inputs (full faithfulness of h^{-1} and a_{fppf}^{-1} with quasi-inverses) are established before the sentence; the same glue step was documented on card L46151 (the ph analogue); unlike the ph version, the essential-image half here uses the fppf-etale sheaf descent lemma rather than an equalizer computation, and no verification is omitted.

*Editorial note.* The label lemma-descent-sheaves-for-fppf-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Comparison of the small etale and big fppf simplicial sites. Same chapter, \label{lemma-compare-simplicial-objects-fppf-etale}. Lines 7304-7306: the diagram h with h_{-1} = a_X and components h_n.
- The fppf-etale comparison lemma. More on Cohomology of Spaces chapter, \label{spaces-more-cohomology-lemma-comparison-fppf-etale}. Lines 7307-7310: (h_{-1})^{-1} fully faithful with quasi-inverse h_{-1,*}, likewise for the components h_n.
- Description of the simplicial-site functor pair h^{-1}, h_*. Same chapter, \label{lemma-morphism-simplicial-sites} (card L46154 of this batch). Lines 7311-7312: componentwise full faithfulness lifts to h^{-1} fully faithful with quasi-inverse h_*.
- Simple sheaf descent for hypercoverings on the big fppf site. Same chapter, \label{lemma-hypercovering-X-simple-descent-sheaves}. Lines 7319-7322: a_{fppf}^{-1} fully faithful with essential image the cartesian sheaves on the big total site.
- Descent of sheaves along a flat, finitely presented, surjective morphism. More on Cohomology of Spaces chapter, \label{spaces-more-cohomology-lemma-descent-sheaf-fppf-etale}. Lines 7340-7345: from h_0^{-1}G_0 =

*Date*: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(a_{0,big,fppf})^{-1}H with a_{U_0} flat lfp surjective, produces F on X_etale with H = (h_{-1})^{-1}F.
- Fully faithful functors and their adjoints. Categories chapter, \label{categories-lemma-adjoint-fully-faithful}. Lines 7353-7355: id is isomorphic to a_* a^{-1} when a^{-1} is fully faithful, opening the quasi-inverse argument.

*Source reference.* Original lemma 07RB: sources/stacks-chow/tags.txt, source lines 9861–9861. Prerequisite pointer invoked at proof line(s) 7353; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0DH5: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7233–7233. Prerequisite pointer invoked at proof line(s) 7306; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0DA8: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5065–5065. Prerequisite pointer invoked at proof line(s) 7321; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0D96: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1273–1273. Prerequisite pointer invoked at proof line(s) 7317; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 09A0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5012–5012. Prerequisite pointer invoked at proof line(s) 7320; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7299; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0DGF: sources/stacks-chow/tags.txt, source lines 16870–16870. Prerequisite pointer invoked at proof line(s) 7313; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0DGL: sources/stacks-chow/tags.txt, source lines 16876–16876. Prerequisite pointer invoked at proof line(s) 7339; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 0DGE: sources/stacks-chow/tags.txt, source lines 16869–16869. Prerequisite pointer invoked at proof line(s) 7310; it names a standard

fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $U$ be a simplicial algebraic space over $S$. Let $a : U \to X$ be an augmentation. If $a : U \to X$ is an fppf hypercovering of $X$, then*

$$a^{-1} : Sh(X_{étale}) \to Sh(U_{étale}) \quad and \quad a^{-1} : Ab(X_{étale}) \to Ab(U_{étale})$$

*are fully faithful with essential image the cartesian sheaves and quasi-inverse given by $a_*$. Here $a : Sh(U_{étale}) \to Sh(X_{étale})$ is as in Section 0DE7.*

*Proof.* We will prove the statement for sheaves of sets. It will be an almost formal consequence of results already established. Consider the diagram of Lemma 0DH5. In the proof of this lemma we have seen that $h_{-1}$ is the morphism $a_X$ of More on Cohomology of Spaces, Section 0DGE. Thus it follows from More on Cohomology of Spaces, Lemma 0DGF that $(h_{-1})^{-1}$ is fully faithful with quasi-inverse $h_{-1,*}$. The same holds true for the components $h_n$ of $h$. By the description of the functors $h^{-1}$ and $h_*$ of Lemma 0D96 we conclude that $h^{-1}$ is fully faithful with quasi-inverse $h_*$. Observe that $U$ is a hypercovering of $X$ in $(Spaces/S)_{fppf}$ as defined in Section 09A0. By Lemma 0DA8 we see that $a_{fppf}^{-1}$ is fully faithful with quasi-inverse $a_{fppf,*}$ and with essential image the cartesian sheaves on $(Spaces/U)_{fppf,total}$. A formal argument (chasing around the diagram) now shows that $a^{-1}$ is fully faithful.

Finally, suppose that $\mathcal{G}$ is a cartesian sheaf on $U_{étale}$. Then $h^{-1}\mathcal{G}$ is a cartesian sheaf on $(Spaces/U)_{fppf,total}$. Hence $h^{-1}\mathcal{G} = a_{fppf}^{-1}\mathcal{H}$ for some sheaf $\mathcal{H}$ on $(Spaces/X)_{fppf}$. In particular we find that $h_0^{-1}\mathcal{G}_0 = (a_{0,big,fppf})^{-1}\mathcal{H}$. Recalling that $h_0 = a_{U_0}$ and that $U_0 \to X$ is flat, locally of finite presentation, and surjective, we find from More on Cohomology of Spaces, Lemma 0DGL that there exists a sheaf $\mathcal{F}$ on $X_{étale}$ and isomorphism $\mathcal{H} = (h_{-1})^{-1}\mathcal{F}$. Since $a_{fppf}^{-1}\mathcal{H} = h^{-1}\mathcal{G}$ we deduce that $h^{-1}\mathcal{G} \cong h^{-1}a^{-1}\mathcal{F}$. By fully faithfulness of $h^{-1}$ we conclude that $a^{-1}\mathcal{F} \cong \mathcal{G}$.

Fix an isomorphism $\theta : a^{-1}\mathcal{F} \to \mathcal{G}$. To finish the proof we have to show $\mathcal{G} = a^{-1}a_*\mathcal{G}$ (in order to show that the quasi-inverse is given by $a_*$; everything else has been proven above). Because $a^{-1}$ is fully faithful we have $\mathrm{id} \cong a_*a^{-1}$ by Categories, Lemma 07RB. Thus $\mathcal{F} \cong a_*a^{-1}\mathcal{F}$ and $a_*\theta : a_*a^{-1}\mathcal{F} \to a_*\mathcal{G}$ combine to an isomorphism $\mathcal{F} \to a_*\mathcal{G}$. Pulling back by $a$ and precomposing by $\theta^{-1}$ we find the desired isomorphism. □